



Computer Network and Information Security
final presentation report

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Executive Summary

On top of the existing network infrastructure (lower layer of OSI model) which only perform the transmission operation, a cryptography operation is needed to ensure that data being transmitted from one end user to another cannot be intercepted, altered, and corrupted along the way, to put it in another words, data needs to be encrypted before being sent, and can be decrypted only by a designated receiver after being sent, while being made sure that other parties cannot and can never decrypt the message. Most cryptographic methods that we are using today are built entirely upon mathematical trust, they are built as mathematical puzzles that, apart from the designed solution to encrypt and decrypt, reserved for the designated sender and receiver, can be solved by other methods, just with unpractical solving time (ranging from decades to thousands of years, for one message, such as brute force). But with the introduction of quantum computer, a new form of computer which utilizes quantum property to dramatically increase computing power, although it is still in experimental process, the improved performance compared to traditional computer is believed to be enough to break through the cryptographic methods that we are using everyday, thus, bringing threat to the modern cryptographic infrastructure, so experts have been preparing a new form of cryptography called “post quantum cryptography” that is designed to prevent force-decryption by quantum computer, while at the same time allowing encryption using the current digital infrastructure that we have, to ensure smooth transition between the old binary digital infrastructure to the new incoming quantum digital infrastructure. Additionally, the “harvest now, decrypt later” threat which is a clear and present threat also encourages the speed of the implementation of the new cryptography to be even faster than anyone has anticipated.

Current cryptographic methods

In order to understand how post quantum cryptography works, it is appropriate to start with how the current cryptography works.

Symmetric cryptography: The shared secret model

Analogy : Imagine you want to send your friend a letter, but you suspect that the mail company or other people along the way of the letter transportation will take a peak at it and know the secret that you want to include in the letter, so you instead lock the message in a chest, send the chest to your friend, and send the key to unlock it using another channel that is more secure than the mail company. This is called

symmetric cryptography, we have one key to encrypt and decrypt the message, shared between sender and receiver.

The global standard is the Advanced Encryption Standard (AES), deployed at AES-128 or AES-256 bits. AES operates on 128-bit blocks of data, passing them through complex structural permutations across multiple rounds (SubBytes, ShiftRows, MixColumns, and AddRoundKey). This creates an avalanche effect where changing one bit randomizes the output. Brute-forcing AES-256 requires astronomical resources, making it secure against classical supercomputers. Likewise with other existing cryptography methods that are still in use within our digital infrastructure nowadays, there is no known algorithm that can crack AES-256 effectively, other than brute force.

Asymmetric cryptography (public-key cryptography)

Analogy : Now, instead of sharing the same key between you and your friend, your friend instead broadcasts a key pattern for everyone to know, including you and corrupted mail company, you use that key pattern to make a key, and then lock a chest with that key, with the letter you want to send to your friend inside, but there is a catch, the broadcast key is a special key, once any chest is locked with that key, it can't be opened with anything, including the key itself, it is a one way key, only lock not to unlock, only one thing can open it, a master key which only your friend has, not anyone else, not even you, so you can theoretically lock a chest with the broadcast key, and can never find a way to open it by yourself. The broadcast key is a public key, and the master key is a private key.

The architect uses pairs of mathematically linked public private key. It relies on the "trapdoor functions", mathematical operations that are easy to process in one direction but virtually impossible in reverse without a specific secret key. The website (as learned in class that even website has a cryptography standard) relies on two primary standards.

1. **RSA** Relies on integer factorization. Multiplying two massive prime numbers is easy but factoring the multiplied number back to its before-multiplied form is computationally infeasible for standard computers

Analogy : By using RSA cryptography, the sender, as computer A, would broadcast a way to mix two colors together, let's just between neon green and blue, that is public key, anyone can know a way to mix two colors together, anyone can mix the color by themselves as well, private key is the formula to separate the color out. Without private key, it is impossible to separate the mixed color into original color, drop by drop.

- 2. ECC (Elliptic Curve Cryptography)** Relies on the Discrete Logarithm Problem. It tracks coordinates across geometric algebraic curves (e.g., $y^2 = x^3 + ax + b$) through point addition and multiplication. ECC delivers the same security strength as RSA but requires significantly smaller key sizes.

Analogy : Imagine a cosmic billiard table which is in a shape of a distorted guitar, with edges of the table in a weird curve and no hole, and there is starting point as the center of the table, now you strike the billiard ball N times, making it bouncing around uncontrollably and stops somewhere on the table, the public key is the final resting point of the ball, the private key is how many times you strike the ball.

Quantum threat. Breaking the trapdoor

Quantum computer present a great risk to the traditional cryptography because of its different methods of computation on a fundamental level, by utilizing the quantum property of quantum physics, which allows it to achieve a far more superior computational power than the traditional computer.

The basic of quantum computing

A traditional computer uses classic bits, which can be either 1 or 0, by connecting a large number of bits together, it makes the computer capable of performing a complex computing operation. But quantum computer's bits, called qbits, can occupy both 1 and 0 state at the same time (using quantum physics's property of superposition, where an element can occupy two state at the same time) making one bits capable of a higher computational power. To illustrate, imagine finding an exit in a maze, classic computer would gradually try each way out, encounter one dead end, go back to beginning, try again, but quantum computer can flood the entire maze at the same time and see which way the water flows out to find out a solution to the maze.

To dive a little deeper, an exact quantum property which the quantum computer can utilize includes : Superposition (existing in linear combination of 0 and 1 simultaneously to hold 2^n states), Entanglement (linking qubits so that a state change in one bit instantly updates another bits regardless of distance between bits) and Quantum Interference (manipulating probability waves to cancel out wrong paths and amplify the correct solution)

How quantum algorithms break encryption

Quantum processors run specialized algorithms that can quickly solve trapdoors

protecting classical networks:

- **Shor's algorithm:** Developed by Peter Shor in 1994, this algorithm uses a Quantum Fourier Transform to solve integer factorizations and discrete logarithms in minutes. It completely undermines asymmetric key cryptography like RSA and ECC, enabling an attacker to derive private keys from public keys.
 - To give a picture, Shor's algorithm turns the path finding problem into a finding a pattern in rhythm problem, imagine a massive sound wave which has a repeating patterns which appear in every large interval, let's say every three billions beat, the traditional computer patten finding technique is like trying to count those patterns, those peak manually, it is infeasible because the time it takes to count all of those peak would be unrealistic, but with quantum compter, the sound wave can be flooded into quantum bits, all beats all at once (using superposition), and with Quantum Fourier Transform, the error or the part of the wave that is not the pattern that we are interested in would cancel out with each other, whilst the mentioned pattern that we are interested in would amplify each other, resulting in the correct answer that we are looking for to only appear. Now, try to imagine if the sound wave is like a mathematical problem from the RSA and ECC algorithms.
- **Grover's algorithm :** Developed by Lov Grover in 1996, this algorithm provides a quadratic speedup for unstructured database searches. It reduces the security depth of symmetric keys by half (reducing AES-128 to 64 bits of security). This vulnerability can be neutralized by upgrading to AES-256, which leaves a strong 128-bit quantum-resistant barrier.
 - To illustrate, imagine a wall with N number of locker, we want to find out the locker which contains the object that we are looking for, but the N number is to high to just gradually walk by and look into each locker one by one, Grover's algorithm is like taking a look into every locker all at once, but we won't find out right away of what locker contains the object, instead, in the beginning, all lockers have equal chance of having an object, but every time the algorithm does the look (look through all locker at the same time), it uses Amplitude Amplification to makes the empty locker slowly cancels each other out and fading away, leaving only the locker we are looking for to be left alone.

- The analogy illustrates how the brute force on the symmetric key can be done by using quantum computer, the searching of brute force, from looking into N lockers, become $\sqrt{N}$ instead. With this, AES-128 becomes vulnerable to the algorithm, but the AES-256 is still safe because even with $\sqrt{N}$ searching time is still infeasible to crack it.

Immediate crisis : “Harvest now, decrypt later”

Back to the corrupted mail company analogy, imagine even with secure locking technique, with both asymmetric and symmetric keys, the corrupted mail company knows that they cannot find a way to crack those chest you’ve been exchanging just yet, so what it does is it makes a replica of each chest that has been passing through, knowing that soon, a welding company would release a new weld rod which allows them to crack the chest, and imagine that the mail company has built a large warehouse, storing thousands of chest replica, prepared to be cracked by the new weld rod.

Threat actors and adversarial nation-states are actively intercepting and archiving large volumes of encrypted digital traffic today. Although this data is protected by classical RSA/ECC layers right now, it is being stored in server repositories. The moment a stable quantum computer runs Shor's algorithm, decades of historical communication will be retroactively decrypted. Information requiring long-term confidentiality (such as intellectual property or defense designs) is already vulnerable if transmitted via legacy public-key protocols.

The solution : post quantum cryptography

- **PQC vs QKD** : Quantum key distribution (QKD) is an encryption technique which uses quantum property to secure the data being sent. PQC consists of software-based, classical mathematical algorithm that run efficiently even on a standard legacy computers with any hardware modifications, in other words, QKD requires digital quantum infrastructure to work, PQC can run in your home computer now.
- **Lattice-based cryptography** : Replaces prime factorization with geometric problems mapped across infinite multi-dimensional grids. It injects controlled mathematical noise into matrix linear equations (Learning With Errors). Finding the original vectors requires solving the Shortest Vector Problem, which resists quantum analysis.

The Concept: A hidden location game played inside a massive, 500-dimensional forest of perfectly gridded trees.

- **The Private Key (The Secret Master Compass):** This is the precise, clean geometric formula of the grid. It acts like a compass that knows exactly how to navigate the 500 dimensions and instantly filter out any radio interference or static.
- **The Public Key (The Book of Noisy Coordinates):** This is a map you publish to the world. It contains millions of coordinate equations inside the forest. However, you purposefully inject a tiny splash of random static (noise) into every single line. To an outsider, it looks like a chaotic wall of corrupted data, but it is secretly anchored to your compass.
- **The Message (The Camouflaged Hiding Spot):** To send a message (like a 1 or 0, a friend selects a random handful of coordinates from your public book, blends them together, and hides the message inside that specific geometric point. They throw a final layer of static over the top.
- **How it Works:**
 - * **The Hacker's Trap:** A quantum attacker steals the message but just sees a massive cloud of digital camouflage. Without the compass, they are forced to manually test trillions of 500-dimensional paths to find the true starting tree (the **Shortest Vector Problem**). They get hopelessly lost.
 - **The Safe Return:** The message reaches you. You apply your **Secret Master Compass**, instantly strip away the noise camouflage, and the clean message is revealed.
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- **Hash-Based Cryptography:** Used for digital signatures, this method links basic one-time signatures into hierarchical structures called Merkle Trees. Because its security depends entirely on the one-way properties of standard cryptographic hash functions, it contains no complex algebraic vulnerabilities.

The Concept: A one-way digital meat grinder structured into a giant security tournament bracket.

- **The Private Key (The Raw Ingredients):** A list of 1,000 completely unique, top-secret phrases that only you know (e.g., "Secret-Phrase-1", "Secret-Phrase-2").
- **The Public Key (The Apex Mush):** You throw each of your 1,000 raw secrets into a cryptographic meat grinder (hash function). It spits out 1,000 unchangeable blocks of "mush." You pair those mush blocks up and grind them together, climbing up a tournament bracket (**Merkle Tree**) until you are left with just **one single, ultimate master block of mush** at the very top. You publish this top block as your Public Key.
- **The Message (The Signed Ledger):** Unlike Lattice (which hides data), Hash-based crypto is used to sign documents to prove they haven't been altered. The "Message" is a contract or digital transaction you want to verify.

- **How it Works:**
 - **Signing the Message:** To sign a document, you take the contract and use "Secret-Phrase-1" from the bottom of your bracket. You grind them together along with the intermediate "proof blocks" (the specific path of mush blocks needed to climb to the top of your bracket). You attach this package to the contract.
 - **The Verification:** The recipient takes your contract, grabs your signature package, and runs it through the meat grinder. They watch the math automatically climb the tournament bracket steps. If the final output matches your **Apex Public Key**, the signature is verified.
 - **The Quantum Trap:** Because grinding data is a strict one-way street, a quantum computer cannot reverse-engineer the top mush block back into your original secrets. It is forced to guess inputs via brute-force, which is easily defeated by using slightly longer mush blocks.

Global Standardization: The NIST Standards

To coordinate a unified defense, the National Institute of Standards and Technology (NIST) finalized three core global Federal Information Processing Standards (FIPS) for production deployment:

Standard Designation	Algorithm Roots	Mathematical Domain	Primary Functional Target
FIPS 203	ML-KEM (Kyber)	Module-LWE Lattice	Key Encapsulation (Replaces RSA, ECDH)
FIPS 204	ML-DSA (Dilithium)	Module-LWE Lattice	Digital Signature Standard (Replaces RSA-Sig, ECDSA)
FIPS 205	SLH-DSA (SPHINCS+)	Stateless Merkle Hash	Backup Cryptographic Signature Scheme

Operational Challenges and Enterprise Migration

Implementing PQC across global legacy networks introduces distinct engineering and operational hurdles.

Key Size Limitations: PQC algorithms require much larger public keys and signatures than legacy protocols. A standard ECC key requires roughly 32 bytes, whereas a post-quantum ML-KEM key exceeds 1,000 bytes. This increase can cause network packet fragmentation during standard handshakes and strain resource-constrained devices like IoT sensors or smartcards.

Migration Strategy: Security environments must adapt using two core principles:

1. **Hybrid Deployments:** Running a dual-layer architecture where payload streams are wrapped in both a classical algorithm (like ECDH) and a post-quantum standard (like ML-KEM). An attacker must compromise both layers to access the data.
2. **Crypto-Agility:** Re-engineering software to eliminate hard-coded dependencies on specific algorithms. Modular abstractions should be used so that cryptographic protocols and parameters can be updated dynamically via configurations without requiring a system rewrite.

Conclusion

While quantum computers offer significant processing potential, they also threaten the cryptographic trust supporting the modern internet. Peter Shor's algorithm undermines asymmetric primitives like RSA and ECC. When combined with active "Harvest Now, Decrypt Later" threat vectors, this challenge requires immediate action. Migrating to hybrid architectures and embracing crypto-agility are essential steps to protect digital infrastructure against the quantum challenge.